School of Future Tech

Case Study Report

on

Stock Market Simulator Game

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1. Introduction to the Case Study

The stock market is an essential part of the global financial system where people invest money in company shares with the expectation of earning profit. However, many beginners are hesitant to participate in real trading because it involves financial risks and complex platforms.

The **Stock Market Simulator Game** is designed to provide a simplified and safe environment where users can learn the basics of stock trading. Instead of using real money, users are given virtual capital and can practice buying and selling stocks while observing how prices change over time.

This project is developed using **HTML, CSS, and JavaScript**, focusing on creating an interactive financial dashboard. The application simulates stock market behavior and allows users to track portfolio growth, profit or loss, and overall investment performance.

The main objective of this project is to help users understand stock market concepts and develop basic investment strategies without facing real financial risk.

2. Problem Statement / Case Background (Abstract)

Many people are interested in learning about the stock market but are afraid of losing money while trading. Real trading platforms are often complicated and require financial knowledge, which can make them difficult for beginners.

This project aims to solve this problem by developing a **Stock Market Simulator Game** that allows users to experience trading in a risk-free environment. The simulator uses virtual money and simulated stock prices to mimic real market conditions.

Through this application, users can:

➢ Buy and sell virtual stocks
➢ Monitor stock price changes
➢ Track portfolio performance
➢ Understand profit and loss calculations

The system helps users gain confidence and basic knowledge about stock market operations.

3. Problem Statement / Case Study Design

The application is designed as a **frontend web-based system** that simulates the process of stock trading. The system contains multiple sections that work together to create a realistic trading experience.

The application includes the following main components:

➢ **Home Page** – Provides an introduction to the simulator and navigation options.

➢ **Market Dashboard** – Displays available stocks with information such as company name, current price, and percentage change.

➢ **Trade Panel** – Allows users to buy or sell stocks using virtual money.

➢ **Portfolio Section** – Shows the stocks owned by the user along with their current value and profit or loss.

➢ **Leaderboard** – Displays rankings based on portfolio performance.

The navigation menu allows users to easily switch between different sections such as **Home, Market, Portfolio, and Leaderboard**.

4. Methods & Algorithms / Technology Applied

The project is implemented using standard frontend web technologies. Each technology plays a specific role in building the application.

HTML – Structure

HTML is used to create the basic structure of the web application. It defines the layout of the stock dashboard, trading panels, portfolio tables, and leaderboard sections. It also organizes different components of the interface to ensure clear navigation.

CSS – User Interface Design

CSS is used to style the application and improve its visual appearance. It creates a professional financial dashboard layout using responsive cards and tables.

Important UI features include:

➢ Color indicators for stock changes (green for gain /red for loss)
➢ Structured and readable data tables
➢ Responsive layout for different screen sizes
➢ Clean and modern design

JavaScript – Functional Logic

JavaScript handles all dynamic features of the simulator. It allows the system to simulate stock price changes and perform trading operations.

Key functionalities include:

➢ **Stock Price Simulation** – Prices change randomly at certain intervals to simulate market volatility.

➢ **Buy/Sell Validation** – The system checks if the user has enough balance before buying stocks and verifies available shares before selling.

➢ **Portfolio Calculation** – The system calculates the total portfolio value and displays profit or loss.

➢ **Local Storage Integration** – User progress and portfolio data are stored using **`localStorage`** so that data is not lost when the page is refreshed.

➢ **Leaderboard Ranking** – Users are ranked based on the total value of their portfolio.

5. Case Study Implementation Details and Snapshots

Application Interface

The Stock Market Simulator Game is implemented as a web-based application that runs directly in a browser. The interface is designed to resemble a simplified stock trading dashboard where users can view market data and perform trading operations.

The main interface includes the following components:

➢ **Header Section** – Displays the title of the application and a reset simulation button.

➢ **Wallet Balance Panel** – Shows the total virtual money available for trading.

➢ **Stock Market Table** – Displays different company stocks along with their current prices and trading options.

➢ **Buy and Sell Buttons** – Allow users to purchase or sell stocks instantly.

➢ **Portfolio Section** – Displays the stocks owned by the user along with their quantity and value.

➢ **Leaderboard Section** – Shows rankings of simulated traders based on their portfolio performance.

The layout is designed using responsive cards and tables so that users can easily understand market data.

Stock Trading Implementation

The trading system is implemented using JavaScript functions that handle user actions and update the interface dynamically.

The main trading operations include:

Buying Stocks

➢ When a user clicks the **Buy button**, the system checks if the user has sufficient wallet balance.

➢ If the balance is sufficient, the stock is added to the user's portfolio and the wallet balance is reduced.

Selling Stocks

➢ When the **Sell button** is clicked, the system verifies whether the user owns that stock.

➢ The system then removes the stock from the portfolio and adds the amount to the wallet balance.

Stock Price Simulation

To simulate real market behavior, stock prices change automatically at regular intervals.

➢ The system uses random number generation to increase or decrease stock prices.

➢ This creates a dynamic environment where users must decide the best time to buy or sell stocks.

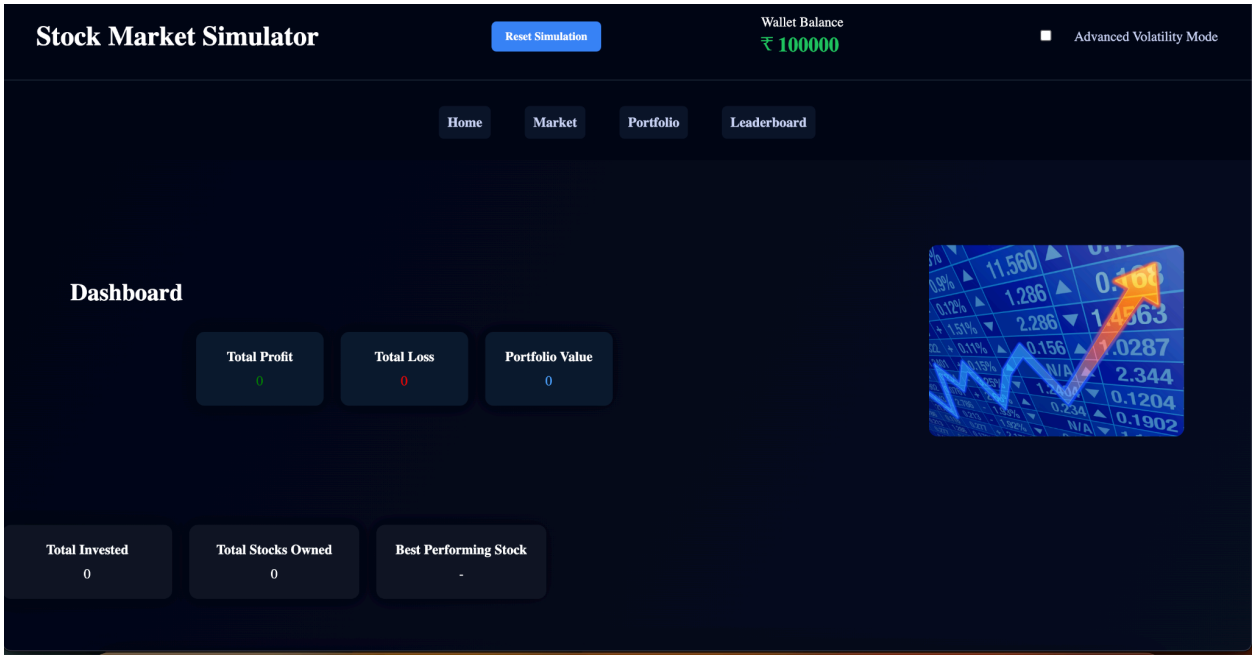
Data Storage

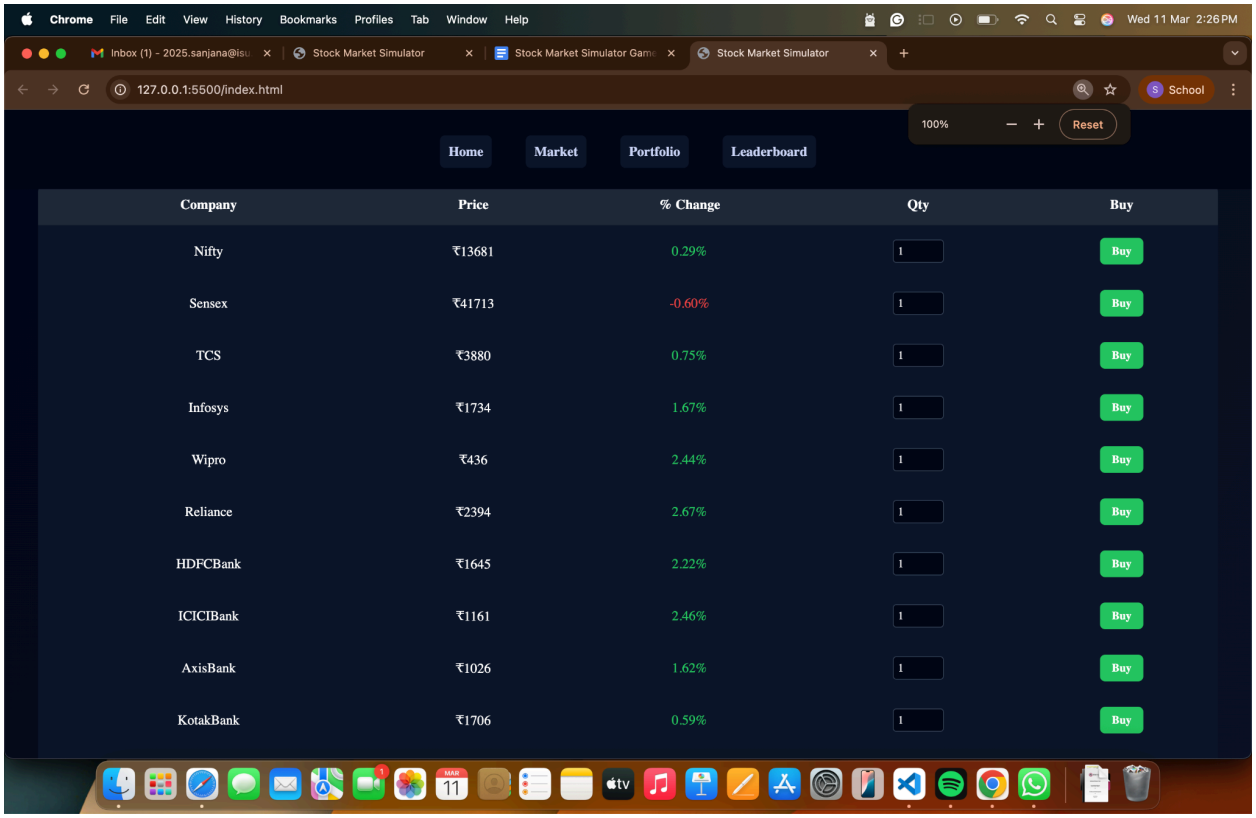
The simulator uses **Local Storage** to save important user data such as:

➢ Wallet balance
➢ Purchased stocks
➢ Portfolio information

This ensures that the user's progress is not lost even if the webpage is refreshed.

Snapshots-





Company	Price	% Change	Qty	Buy
Nifty	₹13681	0.29%	1	Buy
Sensex	₹41713	-0.60%	1	Buy
TCS	₹3880	0.75%	1	Buy
Infosys	₹1734	1.67%	1	Buy
Wipro	₹436	2.44%	1	Buy
Reliance	₹2394	2.67%	1	Buy
HDFCBank	₹1645	2.22%	1	Buy
ICICIBank	₹1161	2.46%	1	Buy
AxisBank	₹1026	1.62%	1	Buy
KotakBank	₹1706	0.59%	1	Buy

Your Portfolio

Stock	Shares	Avg Price	Current Price	Total Value	P/L	Qty	Sell
Titan	1	₹3995	₹3482	₹3482	₹-513	1	Sell
SunPharma	1	₹1444	₹1628	₹1628	₹184	1	Sell
UltraTechCement	1	₹9680	₹9626	₹9626	₹-54	1	Sell

Leaderboard

Rank	Player	Portfolio Value
1	Rahul	₹114590
2	You	₹99276
3	Meera	₹97776
4	Kabir	₹97590
5	Riya	₹92398
6	Arjun	₹87664

6. Case Study Results and Conclusion

The **Stock Market Simulator Game** successfully demonstrates how stock trading works in a simplified environment. The system allows users to experience stock price fluctuations, make investment decisions, and track their portfolio performance.

The project shows that it is possible to create an educational trading simulator using only frontend technologies such as HTML, CSS, and JavaScript. The use of simulated prices and virtual money makes the system safe and accessible for beginners.

Overall, the simulator serves as a useful learning tool for students and new investors who want to understand the basics of stock trading without financial risk.

7. References

➢ W3Schools – HTML, CSS, JavaScript Tutorials
➢ MDN Web Docs – JavaScript Documentation
➢ Investopedia – Stock Market Basics
➢ JavaScript Local Storage Documentation
➢ Figma – UI Design Tool Documentation